Welcome To Your Patient's Personalized Hormone Story

Our intent at EndoAxis is to provide insights and education to help you, the provider, better understand and identify the unique hormonal story that can be unraveled through evaluation of functional hormone metabolite testing.

HOW TO READ THIS REPORT

We have broken down our report into 3 key categories: Overview & Findings, Hormone & Adrenal Insights and Strategy & Analysis sections.

Overview & Findings

The Overview and Findings provides you with a review of the unique hormone and adrenal patterns that emerged through the analysis of 12 urinary sex hormone metabolites and 5 adrenal metabolites and comparing their relationship to one another.

There are over 346 unique sex hormone patterns, only 1 of which will be selected to fit your patient's unique hormone story.

There are 14 adrenal patterns, of which only 1 will be identified for your patient by looking at the unique relationship between free cortisol and metabolized cortisol clearance.

We also include advanced findings from the report that we highlight in the Findings section. These notes are not directly related to the pattern but provide further insights into the unique elements of your patient's hormone results. These are aspects of the report that should not be missed, as they can enhance clinical outcomes.

Hormone & Adrenal Insights

We break down the key hormone relationships in our Hormonal and Adrenal Insights. These sections are intended to provide education regarding the patient's unique imbalances with high-level features to give you detailed insights on the go.

Strategy & Analysis

Our education continues for those who wish to really go deep into the patients' hormonal findings. In our Strategy and Analysis sections, we take every hormone statement made in the report and elaborate further to give you an in-depth understanding of what to focus on for your patient's hormone case.

PREPARED FOR

[redacted]

28 year old female

Cycling - Regular (consistently between 23-35 days in length, give or take 2-3 days)

REQUESTED BY

Micaela Riley

Be Balanced Co.

LABS COLLECTED BY

Dutch Lab

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DISCLAIMER

All comments made on this report are coming from a series of relationships made between the primary sex hormones (progesterone, estrogen and testosterone), which help determine gonadal activity and, in women, ovulatory health. This is coupled with metabolite relationships, which reveal environmental, lifestyle and/or genetic considerations that may be impacting the patients hormone health and can improve hormone outcomes. **For a cycling female, this report requires that the patient collect in their luteal phase of the menstrual cycle, and that any and all bioidentical hormones are provided within the intake questionnaire**.

Nothing can replace the expertise and experience that comes from you, the provider, and your understanding of your patient's unique case and history. As you read through this report, it is ultimately up to you to discern what information is truly relevant and important for your patient's personalized care.

If you are interested in learning more about metabolites, or are curious to learn about disease-specific relationships to hormones, we encourage you to check out our Help Center, accessible through your provider portal and at the end of this report.

Overview

A summary review of the important aspects of your patient's report that you do not want to miss! For further information regarding any highlights, refer to the Educational Section below.

SEX HORMONE PATTERN 18

The patient reports having regular menstrual cycles. Her progesterone is low for an ideal luteal range, while her estradiol is above the 80th percentile of expected luteal range. Make sure to confirm the timing of her test as it relates to her next menstrual cycle. Ideally we want to see 5-9 days between test collection and the start of her next cycle. If this was a mid-luteal sample, she is estrogen dominant and would benefit from follicular support.

Estrogen detox is keeping up with her high estrogen burden. Any concern regarding individual phase 1 metabolite ratios will be discussed in the Findings section below, if applicable.

Testosterone is within an age-expected range.

This pattern reflects estrogen dominance. With regular cycles reported by the patient, this may be an atypical month for her. Ask about any recent illness, stress or drastic change to diet or exercise routine, as these could all transiently influence cycle activity. Errors in follicular development can increase as women enter into perimenopause - resulting in higher surges in estrogen, and less reliable progesterone synthesis. The focus for this pattern is on optimizing follicular and mitochondrial health through the use of targeted phytoestrogenic herbs, minerals and amino acids that will support ovarian health and HPO axis signaling. Bioidentical progesterone cycled in the luteal phase could be considered if it fits with your concerns. Reducing total estrogen burden through support of liver, biliary and GI detox are encouraged, while you work on any factors that could be driving up estrogen production, such as moderate alcohol consumption, stress, obesity, xenoestrogenic exposures or blood sugar dysregulation, to name a few.

Testing serum FSH, LH and prolactin levels on cycle day 3 would be appropriate follow-up, along with a full thyroid panel.

ADRENAL PATTERN 9

Their 24-hour free cortisol average revealed higher free cortisol activity, while their metabolized cortisol is within target range.

This suggests an acute stress picture, wherein the patient can generate a lot of cortisol in response to stress signals, but ultimately is producing and processing an appropriate amount of cortisol during the day. Licorice can slow down cortisol clearance, while elevating free cortisol levels. If they are taking licorice, their dose may need to be adjusted or discontinued.

Keep in mind that high cortisol can have negative sex hormone impact as well.

- For women, higher cortisol can exacerbate menstrual cramping, menstrual flow, PMS, and breast tenderness. It can also increase weight changes, mood changes, and exacerbate acne, hair loss and hirsutism in women of any age.
- In men, high cortisol can exacerbate prostate issues, gynecomastia and weight changes, as well as ED and hair loss.

Support is focused on reducing their acute stress response, if indicated. Some patients will feel great with higher free cortisol, while others can feel more inflamed or anxious. Our targeted formulation blends adaptogenic herbs that help to calm the HPA stress signaling, while amino acids and nutrients help improve adrenal health and cortisol balance.

Findings

A summary review of the important aspects of your patient's report that you do not want to miss! For further information regarding any highlights, refer to the Educational Section below.

ELEVATED RISE IN MORNING CORTISOL

The morning surge in cortisol went above an expected range. This suggests that there was a strong stress signal during their first 2-3 hours of waking. Potential culprits include:

- Waking anxious or stressed
- Low blood sugars - need to eat (gluconeogenesis is kicking in)
- Waking startled (alarm clock, child, pet, noise, etc)
- Anxiety at night - which preprograms the waking response
- Being a natural "morning person"

You do not need to reduce that morning surge if they feel great in the morning. However, if there is a lot of stress in the morning that is impacting their day, or they tend to get very stressed in the evening, which is pre-programing their morning surge, then working on more manageable daily routines, helping them establish healthy boundaries and supporting mind-body medicine may all be considerations.

Formula 13 can help support high-stress states, if indicated.

Hormonal Insights

A high-level review of your patient's hormone interactions.

PROGESTERONE / ESTROGEN BALANCE

Progesterone is low for ideal luteal range, yet her active average estrogens are robust. If she collected at the right time during her cycle, and she is not supplementing with estrogen, testosterone or DHEA, then this pattern indicates that she has ovarian activity, but her corpus luteum was underdeveloped (called luteal insufficiency or a luteal phase defect). This will result in poor progesterone production.

She is estrogen dominant.

High surges in estrogen can occur frequently throughout the month when a woman is approaching perimenopause, as her ovary attempts to release any last remaining follicles, but we can see this in women of any age when there is a higher degree of stress, body fat, blood sugar dysregulation or environmental stress and poor detoxification.

We can also see this pattern emerge if a woman is breastfeeding, while prolactin is suppressing ovulation, yet her ovaries are beginning to restart follicular development.

ESTROGEN DETOX

Her estrogen levels are elevated. When we assess for estrogen dominance potential, we want to look at her rate of clearance through phase 1 and phase 2 detox pathways.

The patient demonstrates poor conversion of estrone into phase 1 metabolites. This means that if you add up all the phase 1 metabolites, they equal less than half of the Estrone total.

- High sugar diets, poor liver health in general, and alcohol consumption and reduce the initial conversion of estrogen into a phase 1 metabolite.
- Fish oils, cruciferous vegetables and increased fiber can support phase 1 detox broadly.
- DIM can support improved estrogen clearance through CYP1A1 and CYP1A2 pathways.

2-OH is within an expected 60-80% target range, which is good.

Phase 2 detox of her catechol estrogens, dependent on methylation, is efficient.

ESTROGEN / TESTOSTERONE BALANCE

Testosterone is within age-appropriate range, while estrogen is elevated. This may be a result of poor estrogen detox, and/or elevated peripheral aromatase activity compounding her ovarian estrogen synthesis.

Although aromatization is a natural process that occurs in the ovaries, extra-ovarian tissues like adipose can increase aromatase expression in response to insulin resistance, inflammation, xenoestrogenic exposures, obesity, leptin resistance, alcohol consumption and even some medications. This can result in an increased conversion of DHEA into estrone, and testosterone into estradiol. Although women do not generate a lot of testosterone, when their ratio

between estrogen and testosterone is skewed toward an estrogenic preference, this can correlate with inflammatory-drive estrogen excess symptoms, such as heavy and painful menstrual periods, weight changes, and fibrocystic breast changes.

Testosterone is metabolized via the 5a or 5b-reductase enzymes. She exhibits a preference for metabolizing testosterone through the 5b-reductase pathway, which is most active in the liver and converts testosterone to 5b-Androstanediol, a metabolite important for bile synthesis but not for tissue use. This preference may indicate a need for rest and digestion, associated with a parasympathetic response, but can exacerbate low androgen symptoms, even if testosterone is within age-range.

For detailed information on hormone synthesis and metabolism, refer to the Strategy and Analysis section of the report.

Adrenal Insights

A high-level review of your patient's adrenal pattern.

We use a weighted average of free cortisol and free cortisone, and here is why:

Free cortisol and free cortisone are both circulating fractions of the same free cortisol hormone. Our body does circulate a low-level of free cortisone that lacks any diurnal curve, but then any produced and circulating free cortisol made by the adrenals will "turn off" in tissues like the gut, sweat glands, salivary glands and kidneys when needed to protect those tissues from mineralocorticoid damage, while tissues like the liver and fat tissue will "turn on" any necessary free cortisone back into free cortisol for anti-inflammatory and stress response as needed, while that recently produced and pulsed cortisol is circulating in the body. This means that any diurnal pattern identified in the cortisone graph had to come from the cortisol that was produced and deactivated at that time, in those tissues.

DIURNAL RISE

The patient has a diurnal response, meaning that as the sun goes up, cortisol goes up, and as the sun goes down, cortisol goes down. Any variation in specific markers throughout the day will be discussed in the Findings section.

ANABOLIC/CATABOLIC BALANCE

Balanced: the ratio of cortisol metabolism is proportionate to the rate of DHEA production.

This does not mean they are within expected range. They may be proportionate, yet still both equally low, equally high, or within target range.

Review the Strategy and Analysis portion of the report for further details on the implications of equally low vs equally high vs equally within target range.

CORTISOL METABOLISM PREFERENCE

Metabolized cortisol favors inactive cortisone, suggesting that their body is "putting on the brakes."

This means their body is favoring less active stress hormones to encourage them to slow down to rest and heal. This can increase fatigue, brain fog, and low mood (to name a few). Supporting a balance of cortisol can be accomplished with herbs like licorice, which support the reduction of inflammation and increase preference for cortisol (THF) if symptoms are present.

PROGESTERONE / ESTROGEN BALANCE

This pattern reflects lower than ideal luteal progesterone production, while estrogen is elevated for an ideal luteal range.

You may want to first confirm the timing of the cycle, as estrogen levels can fluctuate significantly throughout the month. If the patient happened to collect too early in her cycle, we may be capturing too early in her luteal phase, a time when estrogens are just starting to decline following their ovulatory surge, while progesterone has yet to reach peak production. Ideally, she should start her next menstrual cycle 4-9 days after test collection. If her cycle began 10 or more days after testing, her progesterone may actually peak at a higher range. However, even for an early luteal phase, her estrogen is elevated, so improving estrogen activity will be important.

If the timing of the cycle was correct, then the patient is demonstrating a clear estrogen dominant presentation. This can be seen in perimenopause, a time when follicular reserve is declining, while estrogens are surging to attempt to release any last remaining follicles. Perimenopause typically begins around age 42, but can occur as early as age 35 for some women depending on lifestyle and genetics. Women who smoke, for example, are at greater risk for starting their perimenopausal transition earlier in life.

Bioidentical progesterone could be considered if it fits her case to balance her primary sex hormone activity. Often, suboptimal progesterone levels for cycling women occur because of high stress, poor nutrient availability, low body mass, over-exercise, PCOS, or from higher prolactin levels (breast feeding, high stress, prolactinoma, and certain medications like birth control pills, opiates, calcium channel blockers and heartburn medications).

Higher estrogens can occur from high stress, increased alcohol consumption, increased central adiposity, inflammation and high insulin, but may also be seen in the perimenopausal years, as discussed above.

If she is under the age of 45, a product containing Vitex could be considered, with or without the use of bHRT progesterone, to help regulate luteinizing hormone (LH) and progesterone production. Women over the age of 45 may not receive as much benefit from the herbs that help regulate HPO activity, as their ovaries are building a resistance to the signals that the herbs would be supporting.

Supporting detoxification of estrogens will be important, remembering to always work backwards - meaning, start with the gut and biliary conjugation of estrogen, then work on methylation, and finally clean up phase 1 detox if necessary.

ESTROGEN DETOX

Active estrogens, Estrone (E1) and Estradiol (E2) are above an ideal luteal range. Coupled with that, her Phase 1 detox is suboptimal. When estrogen is processed out of the body, it must go through a biotransformation process, taking a cholesterol-based substance (sex hormones) and converting into a water-soluble intermediate for biliary conjugation and clearance out through stool or urine.

Phase 1 detox of estrogen is dependent on a series of enzymes in the CYP450 enzyme family, namely CYP1A1 (the good), CYP3A4 (the bad), and CYP1B1 (the ugly). These enzymes convert estrogen into 2-OH (good), 16-OH (bad) and 4-OH (ugly) metabolites. We expect under ideal circumstances that at least 1/2 of the Estrone in circulation will be processed into phase 1 metabolites, and that there will be a 60-80% preference for 2-OH E1 overall.

For this patient, her phase 1 metabolites are less than half of her total Estrone, but she does maintain an appropriate 2-OH Estrone preference. This suggests that she is sluggish in her initial bio-transformation of estrogen, but has appropriate CYP1A1 enzyme activity to promote 2-OH E1 sufficiently. Cruciferous vegetables, fish oil, and healthy fiber (with a goal of 20-30+ g of fiber daily), can all support estrogen biotransformation.

As her estrogens are elevated, the use of phase 1 enhancers like DIM or I3C may be considered.

Phase 2 detox will expect to follow a similar rule of transformation, that 1/2 of 2-OH E1 will methylate to 2-methoxy E1. In this case, she meets this ratio of conversion, so no further methylation support is necessary.

Sometimes phase 2 metabolites (like 2-methoxy E2) can increase due to poor phase 3 clearance, reflecting that those 2-methoxy E1 metabolites are unable to fully conjugate and eliminate and thus build up in circulation. If she struggles with constipation, bloat, abdominal discomfort, acid reflux, or any other GI symptoms of concern, a GI supportive product to support total estrogen metabolism may be considered.

ESTROGEN / TESTOSTERONE BALANCE

As discussed in Hormone Insights, levels of estrogen are elevated, while testosterone is within expected range for age.

Testosterone should experience a steady decline as women age. Recall that for women, testosterone is produced and secreted by the ovary only to a small degree, and in adipose and adrenal tissue primarily (with about 20-25% of testosterone made and secreted into circulation from the ovary; 25-30% produced and excreted from the adrenal tissue; and 50% produced in peripheral tissues).

Testosterone aromatizes into estrogen through the CYP19A1 pathway (aromatase). For women, this enzyme is very active in the granulosa cells of the ovary and is expected to rapidly convert Androstenedione into estradiol. This enzyme is also present in adipose tissue, the location where the bulk of testosterone is made for women. Although peripheral aromatization of testosterone into estrogen is big a source for estrogen production in women as it is men, there are situations where women could compound their ovarian estrogen with excess adipose estrogen production. Aromatase can upregulate in response to inflammation, obesity, moderate alcohol consumption, high leptin, hyperinsulinemia, and xenoestrogen exposures, as well as in response to some medications like quercetin, white peony and genesteine (soy).

When testosterone is metabolized, it is processed through either the 5a or 5b-reductase pathway primarily (SRD5A1, SRD5A2 primarily). This patient reveals a higher 5b-reductase preference.
5b-reductase is most active in the liver, where it converts testosterone and androstenedione into 5b-Androstanediol. 5b-Androstanediol is considered an active metabolite specifically for bile synthesis, but is inactive for tissue utilization outside the liver.

When a patient favors 5b-reductase activity, it may be in response to a need to rest and digest, with an emphasis on digestion (the dominant parasympathetic response). This is generally not a negative finding, as it suggests that androgens were not driven through the 5a-DHT pathway, and are less anabolic at a tissue level. This can decrease risk for tissue proliferation, oxidation and androgenic symptoms like acne, hair loss and hirsutism. However, a 5b preference does indicate that once testosterone or DHEA are bound to androgen receptors, no further activity is rendered. With very little testosterone to begin with, this could worsen low androgen symptoms like fatigue, low libido, poor endurance stamina, mood changes, poor concentration and poor muscle mass.

Medications like Minoxidil (Rogaine), as well as herbs like saw palmetto, nettle root, pygeum and Reishi mushroom can inhibit 5a-reductase. If they are on any of these substrates, then their results suggest that the substrate is working to reduce 5a activity, and could be continued if they do not struggle with low androgen symptoms. However, if they take these compounds for more superficial reasons (hair loss, acne) and they are symptomatic with low androgen symptoms, then you may want to either reduce their dose or stop the product altogether.

Women can naturally balance and improve their 5a/5b activity by increasing lean protein in the diet, adding in weight-resistance exercises to their daily routine, improving sleep quality and duration, and managing stress (to help the body recalibrate the need to rest and heal). Herbs like maca, tribulus and shatavari can improve symptoms of low androgens, while not directly impacting SRD5a1/5a2 enzymes, which can support symptoms without the risk for unwanted side-effects.

If you were to consider adding testosterone or DHEA bioidentical therapies, they will likely tolerate the hormones well, as they are less likely to convert to higher levels of DHT (a very tissue active intermediate of testosterone). That said, DHEA will push SRD5A transcription, which can increase the 5a-metabolites. For this reason, continue to monitor sex hormone metabolites every 4-6 months after starting treatment, to ensure appropriate metabolism is maintained, and not pushed heavily into the opposite pathway (5a-reductase).

ADRENAL

The 24-hour free cortisol average production is high-end or frankly above range. This reflects their acute stress response at key points during the day but is only representing about 1-5% of total free cortisol (it is a snapshot).

Metabolized cortisol is within an expected target range. Metabolites of cortisol reflect roughly 90% of all free cortisol generated and cleared out of the body on the day of testing. This value suggests that they are averaging an appropriate amount of free cortisol throughout the day.

Higher free cortisol compared to metabolites of free cortisol can occur with high psychological stress, pain, blood sugar dysregulation, or stimulant use. It can also occur if someone is exercising prior to any of their collection points. Higher free cortisol, specifically at the morning point (B point), can be present if someone is naturally a "morning person." If the patient is naturally a "morning person," and can wake up and feel great in the morning, this pattern may be appropriate for them.

Licorice can slow down the rate of cortisol metabolism, extending both the output and circulation of free cortisol. If the patient is actively taking licorice supplementation, their dose should be decreased, or discontinued, while the support of their HPA axis and adrenal tissue recovery is made a priority.

The focus should be on addressing the daily stressors, including stabilizing blood sugars, focusing on restorative movement, and reducing stimulants if needed. Nutrients and herbs that help reduce stress while still supporting total adrenal health are encouraged.

DIURNAL RISE

The patient has an appropriate diurnal pattern, suggesting cortisol rises with the sun, and goes down as the sun goes down. This indicates that there is a brain signal being sent from the hypothalamus (CRH) to pituitary (ACTH), and

down to the adrenal glands AND that the adrenal glands are responding to appropriate signals.

Even if total daily output is low or high, the fact that a pattern is present suggests that the lines of communication from brain to adrenals are intact.

ANABOLIC/CATABOLIC BALANCE

Their DHEA and Cortisol production are in proportion to one another, indicating an equal rise and fall from the adrenal cortex in response to ACTH (from the zona fasciculata and zona glomerulosa).

If they are both equally high, this could indicate a state of heightened inflammation or stress that the adrenals are responding to by releasing ACTH to target the adrenal cortex.

If they are both equally low, this could indicate that the hypothalamus has become blunted to the stress signals received over time and is no longer triggering release of ACTH at the intensity it once would have to stress.

If both levels are within range, and are equally elevating at a similar degree, this is ideal, and demonstrates that when the patient is at rest, the natural adrenal function is to remain balanced in production of these two adrenal cortex-derived stress hormones (DHEA and cortisol).

CORTISOL METABOLISM PREFERENCE

Metabolized cortisol strongly favors inactive cortisone in tissue clearance.

Higher preference for inactive cortisone can be a sign that the body is trying to "put on the brakes" to limit cortisol activity, which could result from chronic stress or a need to dampen cortisol activity globally due to prolonged inflammatory triggers.

Metabolism of cortisol is dependent on the clearance of circulating free cortisol from the body. To become a metabolite, the cortisol had to have been circulating as free hormone at some point during the day. When cortisol is required at a tissue level in active form, the liver or fat tissue can take any circulating free cortisol and keep it active so that it is available to "put out fires." If there is not a strong inflammatory response in the body, then tissues like the colon, liver, kidney, salivary and sweat glands can deactivate the cortisol into cortisone for "safe" circulation and clearance.

Some medications like beta blockers, ACE inhibitors, ARBS, diuretics, benzodiazepines and epileptic meds (including gabapentin), herbs like holy basil and bitter melon, and hyper-thyroid states (or too much thyroid medication) can influence the cortisone metabolism preference as a protective mechanism. Progesterone is a trigger for 11bHSD 2 shifting cortisol into a cortisone preference, so either higher progesterone production, or even progesterone supplementation may push individual into a cortisone state. Elevated estrogens may have a similar impact on this enzyme, and push into a stronger cortisone preference. Patients who frequently meditate or who are very relaxed by nature will often favor inactive cortisone naturally as well, as they simply have less epi-genetic stressors, decreasing the need for higher cortisol reliance. Although not that common to see, if a patient is very balanced and feels great, the inactive cortisone preference may not a concern.

Often a cortisone metabolism preference is seen as a defense strategy to protect our cells from the chronic

degradation and damage that cortisol can have on the tissue when elevated for prolonged periods of time.

ELEVATED RISE IN MORNING CORTISOL

There was a higher rise in cortisol than is generally expected at their morning sample (B). Patients are instructed to collect their morning cortisol sample 2-3 hours after waking up and starting their day. During this time, the instructions are to avoid coffee and to limit physical exertion if possible, in order to achieve a baseline level of cortisol production in a neutral state.

This may be due:

- **Morning anxiety or stress:**
  - waking up feeling startled or panicked, anticipatory stress, exercising between samples, running late or being stressed about work or errands, or having a frantic start to their day morning.
- **Early birds**:
  - This pattern can also be seen in people who are naturally "morning" people. "Morning type" people have been shown to produce a larger surge of cortisol upon waking than "night owl" types. If they wake up and feel great, this rise in cortisol is not a concern.
- **Chronic pre-programming to high stress**:
  - Morning cortisol surges are programed by the amount of stress signaling received by the hypothalamus the day before, so supporting a high morning surge starts by treating high daily stress levels. We do not want to suppress cortisol in the morning, because of its numerous roles in both brain and immune health, specifically from the surge that occurs in the morning.
- **Hypoglycemia**:
  - Cortisol will also spike in response to low blood sugars, in order to stimulate gluconeogenesis by the liver. If your patient struggles with hypoglycemia, they may benefit from eating a protein-rich breakfast in the mornings.
- **Stimulants**:
  - if they had coffee between samples, or went to the gym, this could drive up morning cortisol. Patients taking stimulating medications will also drive up cortisol activity shortly after the drug is administered. Medications include: Adderall, Ritalin, Vyvanse, Luvox, Wellbutrin, Hydrocortisone, Concerta, Provigil, Xyrem (or others in the stimulating class - these are the most common).
- **Exercise**:
  - if they happened to exercise between the two samples, even if just going for a brisk walk, their cortisol can surge as a natural response to movement.

When you seen an elevated morning cortisol value, you may want to ask your patient what happened on the morning of test collection. With such elevated cortisol, there may be a concern that they were waking up anxious or frantic, running late, or dealing with a stressful situation. If they exercised between their two samples, that will cause a natural and healthy rise in cortisol to occur. If they drank caffeine or took any stimulating medications or herbs between the two samples, the result is reflecting their response to the stimulants.

The degree of cortisol rise in the morning is programed by the amount of perceived stress the day before. This means that if they are high in the morning and are naturally more "stressed" individuals, then treating this higher rise in cortisol will start by supporting stress throughout the day, and mitigating the impact of stress on the hypothalamus with nootropic herbs and amino acids.

Morning-type people have been shown to naturally produce more cortisol in the morning, when compared to night-type people. This has been shown to be healthy and appropriate. If your patient wakes up and feels great upon starting their day, this rise in cortisol may be a non-issue.

A high rise in cortisol in the morning is more favorable than a low rise. It is important to remember that a rise in cortisol is a very important. It is from this rise that dopamine and acetylcholine are triggered for production to help with memory, focus and concentration for the day to come. In fact, if there is a lot of presumed stress for the day to come, it has been shown that this will trigger the hypothalamus to increase CRH output the following day, which will amplify the morning cortisol. If the patient has a lot of anticipatory stressors, you may want to focus on mind-body techniques to support their perceived stress, which can help regulate this pattern if necessary.

It is also the rise in cortisol that supports immune function and helps reset the thymus gland each morning, allowing the thymus to regulate self from non-self T cells. People who lack a morning rise in cortisol are seen to have a greater risk for autoimmune disease, and this is the hypothesized reason for that risk.

All of this to say - when we see a surge in cortisol that exceeds expectations in the morning, there may be factors influencing the rise that are modifiable via lifestyle and dietary support, and those should be incorporated if indicated. However, do not directly inhibit that morning rise in cortisol, as our overall health is dependent on the ability to surge cortisol in the morning.